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Image Appearance-Based Facial Expression Recognition

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We investigate facial expression recognition (FER) based on image appearance. FER is performed using state-of-the-art classification approaches. Different approaches to preprocess face images are investigated. First, region-of-interest (ROI) images are obtained by extracting the facial ROI from raw images. FER of ROI images is used as the benchmark and compared with the FER of difference images. Difference images are obtained by computing the difference between the ROI images of neutral and peak facial expressions. FER is also evaluated for images which are obtained by applying the Local binary pattern (LBP) operator to ROI images. Further, we investigate different contrast enhancement operators to preprocess images, namely, histogram equalization (HE) approach and a brightness preserving approach for histogram equalization. The classification experiments are performed for a convolutional neural network

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(CNN) and a pre-trained deep learning model. All experiments are performed on three public face databases, namely, Cohn–Kanade (CK+), JAFFE and FACES.

Keywords: Facial expression recognition; LBP; deep learning; CNN; Inception-v3.

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1. Introduction

The human face image holds information which is useful to determine the characteristics and identity of the face. Face characteristics and identity determination have several applications. These applications include border security, forensic art, surveillance monitoring and biometrics.[1–3] A face image has the demographic and expression information of an individual. Facial attributes can be used to classify face images based on demography and expression. In this work, we focus on facial expression recognition (FER). Typical applications of expression recognition are mental state recognition, human computer interaction and human behavior understanding.

Classification of images based on representative features has been extensively researched.[4–7] Approaches for expression recognition can be divided into two broad categories: first, approaches based on geometric features extracted from face images such as distance between the eyes, etc.; and second, approaches based on appearance which use the raw pixel data such as intensity histogram.[8] In this paper, we perform experiments on face expression databases. First, we evaluate a convolutional neural network (CNN) and use the results as a benchmark for comparison with a pre-trained deep learning model, namely, Inception-v3. Inception-v3 is a state-of-the-art approach for object recognition as determined in the ImageNet large scale visual recognition challenge (ILSVRC).[9]

The paper is organized as follows. In Sec. 2, we describe the appearance approaches used for FER in this paper. Section 3 describes the classification approaches used in this paper. Experimental setup is explained in Sec. 4. Experimental results are given in Sec. 5. Discussion is presented in Sec. 6, and the paper is concluded in Sec. 7.

2. Background

In this paper, we investigate FER using approaches based on image appearance, as shown in Fig. 1. In the context of the appearance of face images, we consider region-of-interest (ROI) images, difference images and local binary pattern (LBP) as shown in Fig. 1. Each of these are explained in this section.

ROI images are obtained from raw images by extracting the face region from raw face images. ROI images eliminate regions of the raw image which are not relevant for FER. Hence, ROI images are created by cropping the original image to remove regions such as hair and ears. We use object detection using Haar feature-based cascade classifiers proposed by Viola and Jones[10] to obtain ROI images. An illustration is given in Fig. 2.

Figure 2(a) is the original image from the Cohn–Kanade (CK+) database, Fig. 2(b) is the image after conversion to grayscale and contrast enhancement. In Fig. 2(c),

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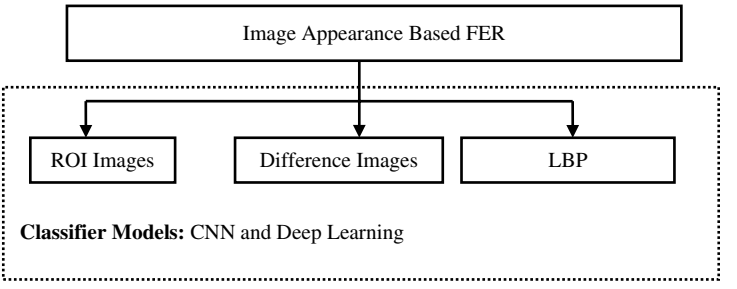


Fig. 1. Overview of FER.

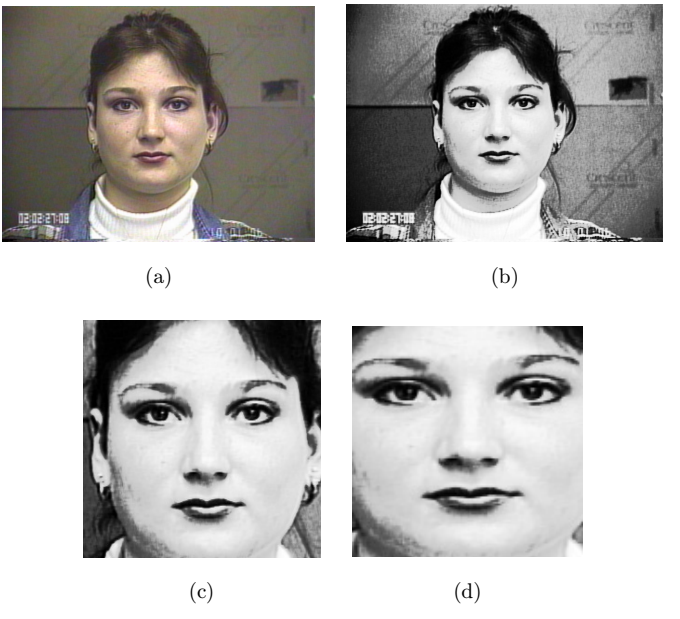


Fig. 2. Preprocessing of face image: (a) original Image, (b) grayscale and contrast Enhancement, (c) crop by face ROI and (d) crop by fixed rectangle and resize.

cropping is applied based on face ROI. Figure 2(d) is the final *ROI image* which is obtained by fixed size cropping and scaling to 128×128 pixels.

The second set of images, namely, *difference images*, are obtained by further processing the *ROI images*. The *difference images* reflect the difference between the neutral expression and the peak expression. Each difference image is obtained from two ROI images, namely, the ROI images for neutral and peak expressions. A difference image is obtained by computing the difference between gray level intensities of corresponding pixels in the ROI images of neutral and peak expressions, as shown in Fig. 3.

(a) (b) (c)

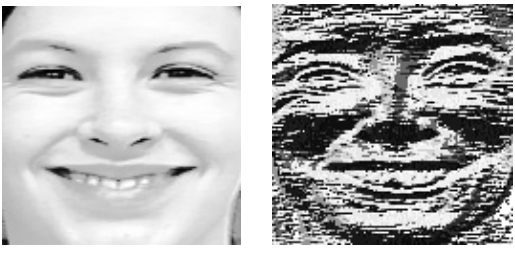
Fig. 3. (a) Neutral expression, (b) peak expression and (c) difference image.

The third set of images are the *LBP images*. The images are obtained by applying the LBP operator to the *ROI images*. Here, we give an overview of LBP.[11] LBP considers the neighborhood of each pixel in the image. It thresholds grayscale values of P neighbors with the center pixel and conducts the code in a binary form. If the grayscale value of the neighbor pixel is greater than or equal to the grayscale value of the center pixel, it is labeled as 1, otherwise it is labeled as 0. The resulting binary values are then converted to decimal and assigned to the center pixel. Each decimal value represents a label to every pixel of the image. The histogram of the labels can be utilized as a texture descriptor. The LBP operator is as follows[11]:

$$\mathrm{LBP}_{P,R}(x_c, y_c) = \sum_{m=0}^{P-1} s(k_P - k_c)2^m, \tag{1}$$

$$s(x) = \begin{cases} 1, & \text{if } x \geq 0, \\ 0 & \text{otherwise.} \end{cases} \tag{2}$$

Here, P is the number of the neighbors, R is the radius of the circle surrounding the center pixel (x_c, y_c), k_c is the gray value of the (x_c, y_c) and k_p is the gray value of the neighboring pixels. The basic LBP operator considers eight neighbors. The operator can be extended to cover different number of neighbors (P) and allow for different radius pixels (R). Hence, $\mathrm{LBP}_{8,1}$, $\mathrm{LBP}_{8,2}$, $\mathrm{LBP}_{16,2}$ are some examples of the LBP operators. An illustration is given in Fig. 4.



(a) (b)

Fig. 4. (a) Raw image and (b) LBP image.

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We considered eight neighbors ($P = 8$) and one-pixel radius ($R = 1$) for the LBP operator. An example of a *LBP image* is shown in Fig. 4(b).

The classification accuracy achieved for ROI images is validated against the classification accuracy for difference images and LBP images. A CNN and a pre-trained model are used as the classifier models for the approaches based on appearance.

3. Classification

We evaluate expression recognition of face images using two classifier models, namely, a CNN and a pre-trained model (Inception-v3).

3.1. *CNNs*

CNNs are the current state-of-the-art model architecture for image classification tasks. CNN extracts high level image features by applying a series of convolution filters to the raw pixels of an image which the model can then use for classification. We use CNN for expression recognition based on image appearance (see Fig. 1). A typical CNN architecture is composed of a stack of convolutional modules; each module consists of a convolutional layer followed by a pooling layer. The convolutional layer extracts features, and the pooling layers down-sample the data extracted by the convolutional layer. The last convolutional module is followed by one or more dense layers that perform classification. The CNN architecture employed is explained in the next section.

3.2. *Inception-v3*

CNN described above requires a significant amount of data and resources for training. Transfer learning approach is used to reduce the training overhead. In this approach, a pre-trained model is used for new domains. Transfer learning models are divided into four categories: instance-based transfer learning, parameter transfer learning, feature-representation transfer learning and relational-knowledge transfer learning.[11] In instance-based-transfer learning, certain parts of the data in the source domain can be reused in the target domain by reweighting. Parameter-transfer learning assumes that the individual models for source domain and target domain share some parameters or prior distributions of hyper-parameters. Feature-representation-transfer learning aims at finding a good feature representation for the target domain. The relational-knowledge-transfer learning deals with the transfer learning problems in the relational domains.

In this paper, feature-representation transfer learning is most appropriate. We use a CNN that has been pre-trained on ImageNet,[12] remove the last fully-connected layer, and then use the rest of the CNN to extract features from the new dataset. Each image is reused multiple times during training. Calculating each bottleneck takes a significant amount of time; these features are used as input to train the last fully-connected layer for the new dataset. The advantage is that the training time is greatly reduced, and the disadvantage is that training accuracy and generalization

ability cannot be guaranteed. We use a pre-trained version of the Inception-v3[13] model architecture because it provides high accuracy results. The improvement of Inception-v3 and Inception-v2 compared with Inception-v1 is achieved from the factorization of convolutions into smaller convolutions. This accelerates the computations, and the extra computing power can be used to deepen the network.

4. Experimental Setup

We performed experiments for expression recognition of face images. In this section, we describe the databases used in the experiments. We also explain the preprocessing of images which is required for expression recognition based on image appearance.

4.1. *Databases*

Approaches for FER based on image appearance (see Sec. 2) are extensively evaluated on three face expression databases, namely, CK+,[14] JAFFE[15] and FACES.[16] Each of these are described below.

The original CK database includes 65% females and 35% males with age ranges between 18–30 years old. The second version of this database which is referred to as the CK+ database has an additional 27% of subjects. It comprises of 123 subjects and 593 sequences. Each sequence has a series of facial images from neutral to a peak expression. The images display seven facial expressions: neutral, sad, disgust, contempt, fear, anger and happiness.

JAFFE database contains 213 images of facial expressions posed by 10 Japanese females. The images display six facial expressions: neutral, sad, disgust, fear, anger and happiness. The images have been rated on the six emotion adjectives by 60 Japanese subjects. The photos were taken at the Psychology Department in Kyushu University.

FACES database contains face images of 171 young ($n = 58$), middle-aged ($n = 56$) and old ($n = 57$) people displaying each of six facial expressions: neutrality, sadness, disgust, fear, anger and happiness. It comprises two sets of pictures per person and per facial expression, resulting in a total of 2052 images. It was developed at the Center for Lifespan Psychology, Max Planck Institute for Human Development, Berlin, Germany.

4.2. *Image preprocessing*

Image preprocessing is required for classification based on the image appearance approach adopted for FER. The CNN classifier is trained and tested on three sets of face images which are obtained by preprocessing the original images in the face databases. The three sets of images obtained by preprocessing are *ROI images*, *difference images* and *LBP images* (see Fig. 1).

4.3. *Contrast enhancement*

A significant step in the preprocessing described above is contrast enhancement. In this section, we address contrast enhancement in further detail.

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Histogram equalization (HE)[17] is a widely used approach for contrast enhancement because of its simplicity and effectiveness. By applying HE, the histogram of pixel intensities is remapped to achieve a better distribution. This improves the contrast in regions of low local contrast. HE accomplishes this by spreading out the most frequent intensity values. The drawback of this traditional approach is that it cannot preserve the mean image brightness because of the remapping of histogram peaks. In order to overcome the drawbacks of HE, several brightness-preserving approaches have been proposed. An effective brightness-preserving approach for contrast enhancement is the brightness-preserving dynamic fuzzy HE (BPDFHE).[18] BPDFHE is implemented in four stages: fuzzy histogram computation; partitioning of the histogram; dynamic HE of the partitions and normalization of the image brightness. In BPDFHE, the histogram peaks are not remapped. Brightness is preserved by remapping the gray-level values in the valley between two consecutive peaks. We investigate FER using HE and BPDFHE for contrast enhancement.

5. Experimental Results

We performed expression recognition of face images in CK+, JAFFE and FACES databases (see Sec. 4). We conducted experiments to investigate classification accuracy based on image appearance. TensorFlow API[19] is used for implementing the experiments. Two sets of experiments were conducted. The first set of experiments were used to evaluate expression recognition using CNN. The second set of experiments were used to evaluate expression recognition using transfer learning model.

5.1. *Classification using CNN*

We performed three experiments in this category. In the first experiment, we used the raw pixel data of face images (see Fig. 2(d)) for training and testing a CNN classifier. A diagrammatic representation of the CNN architecture employed in the experiments is shown in Fig. 5.

The layers in the architecture of Fig. 5 are explained below.

- Convolutional layer #1: Applies 32 5×5 filters (extracting 5×5-pixel subregions), with ReLU activation function.
- Pooling layer #1: Performs max pooling with a 2×2 filter and stride of two (which specifies that pooled regions do not overlap).
- Convolutional layer #2: Applies 64 5×5 filters, with ReLU activation function.
- Pooling layer #2: Again, performs max pooling with a 2×2 filter and stride of two.
- Dense layer #1: 1024 neurons, with dropout regularization rate of 0.4 (probability of 0.4 that any given element will be dropped during training).
- Dense layer #2 (Logits layer): 8 neurons, one for each expression (0–7).

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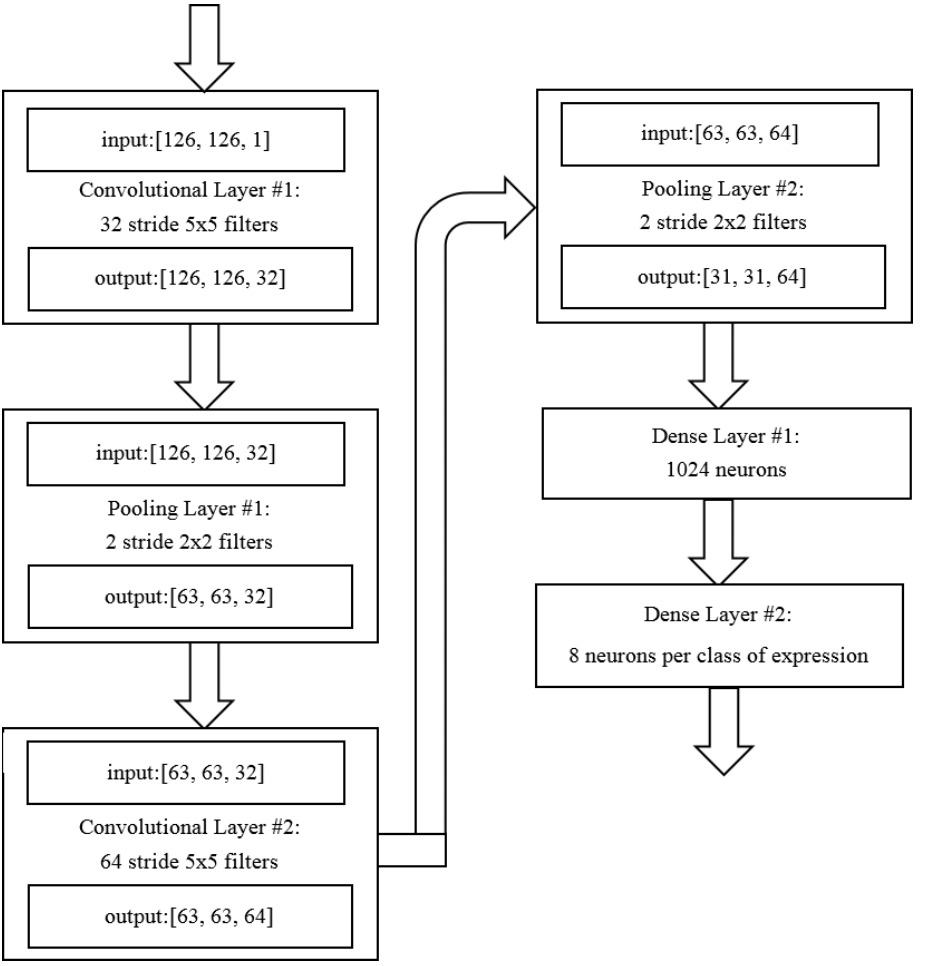

Fig. 5. CNN architecture.

The CNN model was applied to the preprocessed face images (as shown in Fig. 2(d)). In the second experiment, we applied the same CNN model to difference images (as shown in Fig. 3(c)). In the third experiment, we applied the same CNN model to the LBP images (as shown in Fig. 4(b)). The classification accuracy for the three experiments using 10-fold cross validation are shown in Table 1.

5.2. *Classification using Inception-v3*

The process of fully training the model used in the above experiments has high computational cost.[13] Transfer learning is used to avoid fully training the model. We used transfer learning technique by repeating the above experiments with a pre-trained model, namely, Inception-v3.[13] This model is pre-trained on ImageNet.[9]

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Table 1. Expression recognition of images using CNN.

	CK+		JAFFE		FACES	
Image	Hist. Eq.	BPDFHE	Hist. Eq.	BPDFHE	Hist. Eq.	BPDFHE
Raw	73.53	77.03	56.26	54.33	67.38	68.35
Difference	85.19	81.55	65.17	54.28	84.38	79.24
LBP	78.72	79.96	53.51	57.99	78.67	76.21

Table 2. Expression recognition using Inception-v3.

	CK+		JAFFE		FACES	
Image	Hist. Eq.	BPDFHE	Hist. Eq.	BPDFHE	Hist. Eq.	BPDFHE
Raw	76.52	80.45	75.88	68.67	82.19	76.74
Difference	51.65	52.41	51.20	37.41	63.24	57.87
LBP	63.64	53.43	55.88	51.82	68.25	69.91

We retrained the final layer of Inception for FER using *ROI images*, *difference images* and *LBP images*. The results of the experiments are shown in Table 2.

6. Discussion

The FER results for approaches based on image appearance are shown in Tables 1 and 2. The experiments were performed using CK+, JAFFE and FACES face databases. Two contrast enhancement techniques were used — a technique based on the traditional HE and a brightness preserving technique (BPDFHE). The results in Table 1 are obtained by fully training a CNN model. We repeated the experiments for expression recognition (based on image appearance) by using Inception-v3 which is a pre-trained model. Table 2 shows the results obtained by Inception-v3.

Brightness preserving contrast enhancement (BPDFHE) does not improve the results. The simple HE approach for contrast enhancement generally achieves better results. Inception-v3 has the best accuracy for FACES database. This can be attributed to the relatively large number of images in the FACES database; the pre-trained model needs a large database to train for new classes. Between CNN and Inception-v3, there is no clear winner for CK+ and JAFFE databases. The results show that CNN has the best performance for difference images followed by LBP images and raw images. On the other hand, Inception-v3 has the best performance for raw images followed by LBP images and difference images. When using difference images for FER, the pre-trained model has very low accuracy, because the database for training and testing is halved when difference images are computed, because two ROI images are used to generate a single difference image.

7. Conclusion

In this work, we evaluated the appearance approaches for expression recognition of face images. We performed extensive experiments on face expression databases.

First, we evaluated a CNN and use the results as benchmark for comparison with a transfer learning model (Inception-v3). Inception-v3 is a state-of-the-art approach for object recognition as determined in ILSVRC.[9] Generally, deep CNNs have a high computational cost of training because each layer needs to be trained.[13] Transfer learning techniques can be effective for FER when pre-trained models are trained for new classes; however, large databases are needed to effectively train for new classes.

Acknowledgment

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References

1. A. J. O'Toole, P. J. Phillips, X. An and J. Dunlop, "Demographic effects on estimates of automatic face recognition performance," *Image Vis. Comput.* **30**, 169–176 (2012).
2. M. Demirkus, K. Garg and S. Guler, "Automated person categorization for video surveillance using soft biometrics," *Proc. SPIE, Biometric Technology for Human Identification VII* , Vol. 7667, p. 76670P, 2010.
3. A. Dantcheva, C. Velardo, A. D'angelo and J. L. Dugelay, "Bag of soft biometrics for person identification," *Multimedia Tools Appl.* **51**, 739–777 (2011).
4. X.-Y. Jing and D. Zhang, "A face and palmprint recognition approach based on discriminant DCT feature extraction," *IEEE Trans. Syst. Man Cybern. B* **34**(6), 2405–2415 (2004).
5. X.-Y. Jing, X. Zhu, F. Wu, R. Hu, X. You, Y. Wang, H. Feng and J.-Y. Yang, "Super-resolution person re-identification with semi-coupled low-rank discriminant dictionary learning, *IEEE Trans. Image Process.* **26**(3), 1363–1378 (2017).
6. X. You, Q. Li, D. Tao, W. Ou and M. Gong, "Local metric learning for exemplar-based object detection," *IEEE Trans. Circuits Syst. Video Technol.* **24**, 1265–1276 (2014).
7. Z. Zhu, X. You, C. L. P. Chen, D. Tao, W. Ou and X. Jianga, "An adaptive hybrid pattern for noise-robust texture analysis," *Pattern Recognit.* **48**(8), 2592–2608 (2015).
8. J. Anil and L. P. Suresh, "Literature survey on face and face expression recognition," in *Proc. Int. Conf.Circuit Power Computing Technology* (IEEE, 2016), pp. 1–6.
9. O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein, A. C. Berg and L. Fei-Fei, "ImageNet large scale visual recognition challenge," *Int. J. Comput. Vis.* **115**(3), 211–252 (2015).
10. P. Viola and M. Jones, "Rapid object detection using a boosted cascade of simple features," in *Proc.Computer Vision and Pattern Recognition*, Vol. 1 (IEEE, 2001), pp. 511–518.
11. A. Suruliandi, K. Meena and R. R. Rose, "Local binary pattern and its derivatives for face recognition," *IET Comput. Vis.* **6**, 480–488 (2012).
12. S. J. Pan and Q. Yang, "A survey on transfer learning," *IEEE Trans. Knowl. Data Eng.* **22**(10), 1345–1359 (2010).
13. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition* (IEEE, 2016), pp. 2818–2826.

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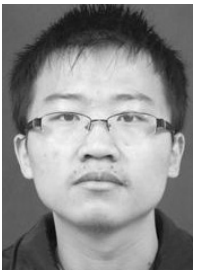
14. T. Kanade, J. F. Cohn and Y. Tian, "Comprehensive database for facial expression analysis," in *Proc. 4th Int. Conf. Automatic Face and Gesture Recognition* (IEEE, 2000), pp. 46–53.
15. M. J. Lyons, S. Akamatsu, M. Kamachi and J. Gyoba, "Coding facial expressions with gaborwavelets," *IEEE Int. Conf. Automatic Face and Gesture Recognition* (IEEE, 1998), pp. 200–205.
16. N. C. Ebner, M. Riediger and U. Lindenberger, "FACES — A database of facial expressions in young, middle-aged, and older women and men: Development and validation," *Beh. Res. Meth.* **42**(1), 351–362 (2010).
17. R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, 2nd edn. (Prentice Hall, 2002).
18. D. Sheet, H. Garud, A. Suveer, J. Chatterjee and M. Mahadevappa, "Brightness preserving dynamic fuzzy histogram equalization," *IEEE Trans. Consum. Electron.* **56**(4), 2475–2480 (2010).
19. M. Abadi, A. Agarwal, P. Barham, E. Brevdo, Z. Chen, C. Citro, G. S. Corrado, A. Davis, J. Dean, M. Devin and S. Ghemawat, arXiv:1603.04467.



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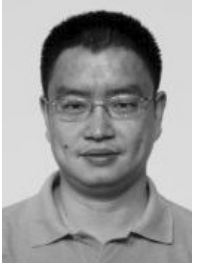

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